

TREATMENT
OF
CHOLERA

IN

The Royal Hospital, Haslar,

DURING THE MONTHS OF JULY AND AUGUST, 1849,

WITH

REMARKS ON THE NAME AND ORIGIN
OF THE DISEASE.

BY

JOHN WILSON, M. D., F. R. S.,

INSPECTOR OF NAVAL HOSPITALS AND FLEETS.

SIMPKIN, MARSHALL, AND CO., LONDON;
LEGG, GOSPORT.

1849.

TREATMENT
OF
CHOLERA

IN
The Royal Hospital, Haslar,

DURING THE MONTHS OF JULY AND AUGUST, 1849,

WITH
REMARKS ON THE NAME AND ORIGIN
OF THE DISEASE.

BY
JOHN WILSON, M. D., F. R. S.,
INSPECTOR OF NAVAL HOSPITALS AND FLEETS.

SIMPKIN, MARSHALL, AND CO., LONDON;
LEGG, GOSPORT.

1849.

The Project Gutenberg eBook of Treatment of Cholera in the Royal Hospital, Haslar

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Treatment of Cholera in the Royal Hospital, Haslar

Author: John Wilson

Release date: December 29, 2021 [eBook #67041]

Most recently updated: October 18, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/67041

Credits: Transcribed from the 1849 Simpkin, Marshall, and Co. edition by David Price. Many thanks to the British Library for making their copy available

*** START OF THE PROJECT GUTENBERG EBOOK TREATMENT
OF CHOLERA IN THE ROYAL HOSPITAL, HASLAR ***

Transcribed from the 1849 Simpkin, Marshall, and Co. edition by David Price. Many thanks to the British Library for making their copy available.

TREATMENT
OF
CHOLERA
IN
The Royal Hospital, Haslar,
DURING THE MONTHS OF JULY AND
AUGUST, 1849,
WITH
REMARKS ON THE NAME AND
ORIGIN
OF THE DISEASE.

BY

JOHN WILSON, M.D., F.R.S.,

INSPECTOR OF NAVAL HOSPITALS AND FLEETS.

SIMPKIN, MARSHALL, AND CO., LONDON;
LEGG, GOSPORT.

1849.

TREATMENT OF CHOLERA.

ON admission the patient was immediately placed in a hot bath, from 104° to 112° of Fahrenheit, in relation to the reduction of circulatory power, and of superficial heat; diligent friction of the abdomen and extremities, according to the place and violence of the spasms, being at the same time practiced. The measure seldom failed, except in cases of extreme collapse, to excite some warmth, abate cramps, and lessen suffering, at least, for a time.

While in the bath, or as soon as carried from it, a drachm of tincture of opium, in aromatic water, was administered. If, as almost invariably happened, the draught was instantly rejected, the proper practice of the hospital was forthwith commenced and steadily pursued. It consisted of the following means:—

Two grains of calomel, in bolus, every hour, or every half hour.

Half a drachm of oil of turpentine, in two ounces of mucilage, repeated every hour, or every second hour; most frequently the last. In one instance it was given every half hour.

An enema, consisting of two ounces of oil of turpentine, two drachms of tincture of opium, three ounces of mucilage, and three ounces of camphor mixture, repeated according to circumstances.

Friction applied to cramped parts assiduously, and as forcibly as could be borne, with turpentine and olive oil.

Such, in brief, was the treatment adopted and relied on, though various auxiliaries were, from time to time, and in different cases, employed.

Occasionally, but chiefly when the patient represented the vomited fluid as being sour, a solution of carbonate of soda in water was substituted for plain

water, as a portion of drink. Sinapisms were sometimes applied; aromatics were now and then, though seldom, prescribed; and in two cases, where with moderately firm pulse, spasms were universal and excessively severe, blood was taken from the arm. Pans with hot water were often applied to various parts of the body, but could seldom be long borne, or kept in place, from spasms or other causes of restlessness in the patient.

One remarkable and gratifying effect of the turpentine draughts consisted in what may be called their acceptability to the stomach. While aromatic and cordial mixtures were instantly rejected, they were generally retained for considerable periods. The turpentine injection had similar effects on the intestines, allaying irritation and checking discharges. With few exceptions they arrested the flux for a time, in most instances for a considerable time. So powerful, in fact, was their restraining power, that they did not require to be often repeated. Their controlling influence was unquestionably great.

Many years ago, while serving in the West Indies, and often looking anxiously but in vain, for the constitutional effects of mercury in the precipitous fever of that region, the writer was led to inquire whether some accessory agent might not be found to accelerate and determine the action of the mineral; and turpentine, from its penetrating properties, rapidly entering the circulation, exciting the capillaries, and stimulating the kidneys, presented itself as probably possessing the desired qualities. It was tried, and did not altogether disappoint expectation. It was thought afterwards that it did not only precipitate mercurial action, but increased its remedial power. A paper on the subject was printed in the "London Medical and Physical Journal," especially in reference to the treatment of neuralgic affections, in 1830.

Calomel in some shape, in various quantities, after various intervals, alone or combined, has long been the most popular remedy for cholera; and, from the concurrent testimony of many witnesses, it has properly gained its reputation. The practitioner has not always, perhaps, considered very carefully the grounds on which he prescribed, nor the channel through which he expected its remedial agency, being satisfied with the result. It is certain, however, that the result from it—as from other things—has been too often the reverse of satisfactory. It is equally certain, that if an auxiliary accelerating agent is a desideratum in the precipitous fevers of the West

Indies, it is much more so in dealing with the yet more precipitous disease under consideration. Looking at the matter in this light; believing that calomel, as it is commonly administered, in the worst and most suddenly fatal cases of cholera, seldom passes beyond the stomach; and being satisfied that to act as a remedy it must enter the circulation, and reach secreting extremities, the method of treatment specified above was adopted.

The results on the whole were satisfactory. Though not such as were desired or even hoped for, they at least exhibited a full average amount of success, inspired confidence, and tended to support the belief, that the practice was founded on right principles.

Of 37 cases admitted 12 terminated in death, the remainder in complete recovery. ^[9]

All the cases in the above number were considered to belong strictly to the epidemic cholera of the season, characterized chiefly by depression of vital power, suppression of biliary and urinary secretion, and great tendency to death. Pains were taken to exclude from it allied affections, especially bilious, or as it is sometimes called, English cholera, of which there were many, and some grave cases. Want of care in this respect renders useless comparisons of the respective value of different modes of treatment; and it is suspected that it is more to such carelessness, to use no strange word, than to superior skill in the practitioner, that the high proportion of cures claimed in some instances should be ascribed. This is said without meaning to insinuate that one method of management is not better than another, or wishing to damp the inquiry in which so many men are now anxiously engaged, as to how more may be done than has yet been effected for choleral patients. In attempting to balance the respective merits of different lines of practice, it is also necessary, in order to render the comparison fair or instructive, to know whether as a whole, the cases treated by each were equally severe.

Of the 12 fatal cases which occurred here, eight of the subjects were in a state of complete collapse—cold livid and pulseless—on admission; in another collapse was nearly complete, and death speedily followed in all, without the slightest sign of re-action. In one of the remaining three cases, there were slight and transient periods of re-action, alternations of promise

and discouragement for fifty hours, when fatal sinking came on. In the other two, the first danger was past, but severe re-actionary fever followed. In one there was restoration of the biliary secretion; in the other, during the last eighteen hours the subject had the complete appearance of a patient in West Indian fever, discharging largely from the stomach, and more sparingly from the bowels, a fluid exactly resembling *black vomit*.

In the cases, which terminated in recovery, the impression was severe, though not equally so; and in each the symptoms, as already stated, were considered clearly characteristic of the prevailing epidemic. In some of them no hope was entertained for a time; especially in two cases, where there was in excess, lividity of surface, cold sweats, corrugated skin, bent fingers and toes, and failure of pulse.

In considering the probability of recovery from cholera, there is reason to think that the manner of attack should be taken into account, jointly with the severity of subsequent symptoms. From what was observed here, it appeared that when there had been precedent diarrhœa, or when there had been—though sudden—a gradual progress to the collapsed state, there was a much better chance for the patient than when the disease, in overwhelming force, fell upon him at once. When, soon after eating a hearty meal, in perfect health, the subject is obliged to be relieved from duty in the ranks, or on deck, becoming in an instant faint and giddy, with a rush of fluid from the stomach and bowels, shrinking of features, fluttering pulse, coldness of surface tongue and breath—struck down, as it were, by electricity—to which soon followed the up-turned ecchymosed eye and whispering voice—when the disease thus sets in, it is doubtful whether art has any power to arrest, or materially modify its fatal career. Such, at least, is the impression from what was observed here; and such, without questioning what has been alleged to have been done by others, or disparaging the means they employed, it is apprehended will be the conclusion of most observers elsewhere.

The practice pursued in Haslar Hospital is submitted to the profession, not because it had any very eminent success, nor on account of its including new remedies, but because the proportion of recoveries was at least fully as large as that which has followed other modes of treatment; because there was some novelty in the combination of the means employed; and because

it is thought that any contribution to the therapeutics of cholera will be acceptable.

Among the diseases allied to cholera which have been treated in Hospital during the last two months, should, it is believed, be included grave cases of fever, with striking predominance of gastric symptoms, and excessive discharges from the alimentary mucous surface, as well as the following:—

Forty-three cases of febrile diarrhœa, with rice water digestions, and strong choleral tendency.

Ten cases of colic, with spasms of extremities.

Twenty cases of bilious cholera, making a total, exclusive of fever, of 73 cases of allied affections, all of which have ended in cure, or are making favorable progress. Some consideration of these cases, and of the various appellations applied to cholera, have led to the following remarks on

THE NAME OF THE DISEASE.

Although the term cholera, when applied to the disease under notice, is derivatively erroneous, it has been so long adopted, and universally employed, that it would be vain, perhaps useless, to attempt to alter it; but the adjective appellatives coupled with it are so numerous and inappropriate, so confusing, and so likely to lead to unjust conclusions, that it is desirable to substitute for them a single significant epithet.

Asiatic cholera is the most common designation, and appears to be the most incorrect, inasmuch as it assumes that the disease is an import from Asia, while there is strong reason to conclude that it has no more claim to be called Asiatic than American. There is reason to conclude that the cholera now widely diffused over the United Kingdom, is not an imported product of any foreign country, near or remote, but is as much the product of the places and the circumstances of the subjects where it exists, as is ague, or bronchocele.

Spasm is not peculiar nor essential to it; for severe spasms often accompany bilious cholera; and in the worst forms of this disease cramps are not violent nor continued. Spasmodic cholera, consequently, is not a proper designation.

The terms “malignant,” “pestilential,” &c., which have been joined with it, convey no idea but that of destructive force, and are destitute of discriminative meaning.

With a view to getting rid of these, and such erroneous or unmeaning terms, it is proposed to couple with the substantive *cholera* the adjective *abiliosa*, the prefixed privative being intended to denote the suppression of the biliary secretion, so constant and important an element in the diseased actions constituting cholera. It is true that other secretions essential to health, as that of urine, are suspended, or materially lessened; but the total

want of bile in the fluid discharged from the stomach and bowels, is one of the most striking and unequivocal characteristics of the disease; and the term suggested would serve to separate it from the form of cholera in which the biliary secretion is excessively augmented, and with which it is perhaps sometimes confounded. It would, at any rate, have the merit of giving one distinctive idea, and leading to no false conclusion. There would then be two intelligible names for the two forms of the disease, namely, *cholera abiliosa* and *cholera biliosa*.

CAUSE OF THE DISEASE.

It is not intended in this place to discuss at length the question of the contagious power, or personal communicability of cholera, but in support of the opinion given above, that it is a domestic, not a foreign malady, with which we have to deal, a few incontrovertible facts will be cited.

The disease broke out at the same time in Gosport, and in Portsea Portsmouth and Southsea, situated on opposite sides of the harbour, and affected numbers in different parts of those towns at once.

At the same time—almost to an hour in some of the places—it appeared in Southampton, Salisbury, Bristol, and Plymouth. Such simultaneous eruptions of disease, in different distant places, appear incompatible with the hypothesis of contagion—irreconcilable with the belief that it arises from, and is communicated by one body to another, either directly or indirectly, either by recent emanations from a diseased body acting speedily on proximate healthy bodies, or by the same emanations, in a concrete form—called formites—acting on distant healthy bodies, after uncertain lapses of time.

In the Minden, hospital ship at Hong Kong, in 1843, when periodic fever and flux were prevalent and highly fatal, a man, convalescing from an attack of the latter, was suddenly seized with unquestionable symptoms of cholera, which ran its destructive course in a few hours. The case is noticed in the “Medical Notes on China,” as curious from its isolation; and a conjecture was hazarded at the time as to some affinity between its cause and that of fever and fluxes, then rife at that place. No other case of cholera appeared before or after it, though the subject was affected and died in the midst of a mass of men, accumulated between the decks of a ship.

A similar case occurred in the Rattlesnake, while employed in the West Indies, in 1826, with this difference, that it happened in a healthy ship to a

healthy man. With incessant rice water vomiting and purging, rapid failure of circulatory power, lividity of surface, cold sweats, cold breath and tongue, the subject sunk in eighteen hours. No other case occurred in the ship, nor was another heard of on the station.

Whatever difficulty there may be in accounting for the occurrence of those cases of cholera, it can scarcely be imagined that they arose from human contagion. It is certain that the disease was not propagated by them.

In the year 1832, the north of Ayrshire generally suffered severely from cholera, while the south part of the county entirely escaped. The disease was excessively fatal in the county town—Ayr—situate about a mile north of the river Doon. It approached close to the north bank of the stream, but did not cross it. From that river to the river Stinchar, a distance of about 30 miles south, no case but one was known to exist; and that one occurred in a letter carrier who had been in Ayr, when, or immediately before he was attacked. He returned to his residence in Girvan, where he soon died with unequivocal symptoms of the disease which was raging in the town from which he came, but had not till then appeared in the town to which he returned. Yet with him the disease not only began, but terminated in Girvan—a poor place, in which the inhabitants are not remarkable for cleanly or orderly habits. Here then was a district extending 30 miles in one direction, by about 20 in the other, bounded on the north by the river Doon, on the west by the sea, on the south by the river Stinchar, and on the east by a chain of hills, where there was but one case of cholera, and that one carried into it, from a deeply affected place 20 miles distant. ^[17]

How was its immunity to be accounted for, if the disease which was destroying so many in Ayr was endowed with a contagious property? The intercourse was uninterrupted, and the district in question populous, containing many villages and considerable towns, including Girvan, with a population of about 5,000, mostly hand-loom weavers, a great majority of whom lived in crowded, ill-ventilated, ill-kept rooms. Cholera was carried there: the place and persons seemed especially fitted to foster and extend a contagious disease; yet the disease made no way there.

These instances, and hundreds of similar import which might be cited, seem to show that cholera is not primarily and necessarily a self-propagating

disease. The question, as to whether, when, and how it has contagious power grafted on it, is one of more difficult solution, in which it is not meant now to enter; but reference may be made briefly to some circumstances which have been alleged in proof of the contagious property of cholera—of power possessed by it, either originally belonging to, or engendered by it.

In the autumn of 1833, Beith, a considerable town in the north of Ayrshire, was suddenly affected by cholera, which, in a few days extended to many persons, and in a few weeks cut off a large proportion of its inhabitants. About the time that the disease broke out, a poor family had arrived from Glasgow, where cholera still lingered, where it had existed more than twelve months, and where during the previous year it had been prevalent and destructive. The poor family that went thence to Beith were not affected by cholera, nor was it shown that they had been in communication with choleral patients in Glasgow. All that was proved against them was the fact that they had lived in that city, and yet on them was charged the introduction of cholera into Beith, and indirectly the mortality which followed.

It is worthy of remark, in illustration, with many such cases, of the mysterious movements of the cause of cholera, that Beith in the preceding year had wholly escaped its power, while adjacent towns and villages with which it had constant intercourse were suffering severely from it. That Beith should evade the contagious power of cholera in 1832, when the disease was rife in its near neighbourhood, and fall under it in 1833, when there is no evidence of its being nearer than Glasgow in mitigated force and occurring rarely, is, to say the least, difficult to understand.

In the spring of this year a custom house officer who had been on board a foreign vessel in the Thames, in which were cases of cholera, was soon after attacked by the disease, and died at Gravesend. A little later, a nurse who had attended a choleral patient in the Dreadnought Hospital ship was attacked by the disease and died. ^[19]

These two cases have been considered by some as all but decisive of the question; they have been looked on as furnishing cumulative proof of the self-propagating power of cholera. But to satisfy others, especially those

gifted with only a moderate share of credulity, it would be necessary to show that there was not at the time an endemic cause of cholera on the banks of the Thames, which, though then but sparingly developed, was capable of exciting the disease in persons strongly disposed to it, by previous disease, destitution, or other debilitating agents. It would be necessary to show that one of the laws of other febrile endemics, such as yellow fever, does not influence this, namely that when the *essential* cause is in much force it attacks persons, though with a certain relation to individual susceptibility, with various degrees, up to the least conceivable degree of it; while reversely, when the cause is little diffused or concentrated, only those who are especially disposed to it—only those who have excessive susceptibility, natural or acquired—constitutional disposition, or disposition from circumstances of life—suffer from it. It can scarcely be denied, that on such difference mainly, if not entirely depends the difference in the prevalence of many febrile endemics; in one instance causing many, in another few attacks; now leading to a sweeping epidemic, and then giving rise to a few cases, or to one case. It would, in short, be necessary to show that while we are trying to trace the disease from one person to another, its cause is not springing from under our feet, and mingling with the air we breathe; which in the cases in question would not be an easy task, seeing that the disease had much endemic extension both before and after their occurrence, and that for years sporadic cases have been reported in the same or neighbouring localities. Nor will the doubts left in the minds of some enquirers, after full consideration of those cases, and allowing them all the weight they deserve, be lessened by reflecting on those which happened in the Minden and Rattlesnake respectively.

But the case of the nurse in the “Dreadnought,” suggests questions respecting the fate of nurses attending choleral patients in other places. If it be true, as it is believed to be, that they are affected by cholera in a degree scarcely, if at all, exceeding that of persons in the same social condition otherwise employed, the fact would appear to furnish a strong general argument against the contagious nature of the disease. How does it happen that persons living in the same room with, and constantly handling and helping patients in all stages of the disease, so constantly escape its power, if it be not only readily communicable by the person to healthy persons,

who are in juxtaposition with its subjects for a short time, but capable of being propagated by their clothes, in distant places, after undefined periods?

Five fresh nurses were brought into Haslar Hospital to attend the cholera patients. They passed at least half their time, night and day, in the ward, and slept there in their turn. They were much employed in rubbing the patients, and consequently leaning over them, administering injections, removing ejections, &c., in short performing all the duties of their place, which were at once trying, laborious, and likely to excite apprehension,—yet none of them was affected.

These, and such facts, of which a multitude might be accumulated, may be objected to, on the ground that they afford negative evidence only, and that one positive proof of contagion would outweigh them all. Be it so. But where is the positive proof to be obtained, and by what distinguishing mark is it to be recognised? Suppose one of the five new nurses brought into Haslar to attend on the subjects of cholera had been attacked by the disease, during, or soon after that service, must it be admitted that the disease was communicated by the patient to his nurse? It is submitted that it should not. Before such admission can be held necessary, it must be shown that the disease which attacked the nurse was not derived from places outside the Hospital, which he was allowed to visit, or from the cause diffused in the air of the neighbourhood, in less or greater concentration, and manifesting its powers relatively to the degree of concentration, and the force of pre-disposing co-operating agents.

It would be unsafe to assert that cholera can never, under any circumstances become contagious; but if the conditions which lead to its acquiring that property, the crisis by which it is effected, and the period of its accomplishment cannot be ascertained, the policy of searching for it may be questioned. Truth is desirable on its own account certainly, but the truth or falsehood of a position is relatively important according to its bearing on the business of life; and if the position that cholera may become contagious could be demonstrated, to what practical purpose could it be turned? It does not appear how it could be used to stay the progress or mitigate the power of the disease. It would not likely be proposed as a reason for separating the sick from the healthy, or enforcing quarantine regulations of any kind. He must be a very ardent believer in the self-propagating power

of cholera who would urge authority to shut up, and surround with guards, houses standing in many different parts of Gosport, Portsea, Portsmouth, and Southsea, and half the towns and villages of England. There is little danger of such measures being adopted now, although they were seriously put forth by the General Board of Health, under sanction of an order in Council, in October, 1831; but, if they were put in force, it would not be rash to assert that they would aggravate the evil immensely. They would turn fear into terror, and interfere with, or prevent the ordinary offices of humanity; thus, at the same time, supplying one of the most powerful predisposing causes of the disease, and fearfully augmenting the misery and danger of the affected. In such a state of things it would be impossible to persuade persons of character, like those who, without hesitation, undertook the duty of nursing the choleral patients in Haslar, to enter on similar service for eighteen-pence a day; nay, it might be impossible to persuade men to enter on such a forlorn hope by any consideration.

Although the pro-contagionist might not be moved by facts and inferences like these, he would perhaps be disposed to ask the man propounding them, some such question as the following.—As you do not admit that contagion is the cause of cholera, what in your opinion is the cause? In return it would not be impertinent to say to the querist, what is the cause of ague? Something emanating from the surface of the earth, on the spot, or not far from the place where its subjects are resident?—something so subtle as to be imperceptible by the senses, and hitherto beyond the scrutiny of chemists and meteorologists. He would, it is presumed, answer in some such terms, or if he attempted any much more precise and instructive, he would proceed without the warrant of fact and experience. Such at least would be the general reply of the profession, and a more complete and unconditional confession of ignorance of every thing respecting the essence of a most powerful and wide spread cause of disease cannot be made. Though we know a good deal of the places most prolific in its production, as well as of the agents which co-operatively increase its force, we know it solely by its effects.

In very similar, if not precisely the same terms, it seems to the writer, should the question, what is the cause of cholera, be answered. Little, if anything more can be affirmed respecting its origin, but almost every thing authentic in its history, progress, and phenomena, testify to its having a

local source, and generally very limited scope of operation. Thus it appears simultaneously in different, distant places, leaving intermediate places untouched. It attacks in one town a particular district, street, or portion of street, beyond which it does not travel. In another town it shows itself among distant portions of the inhabitants, leaving long spaces unscathed. Again, while one town suffers severely from it, another in the neighbourhood has not a single case then, or thereafter; or the town which escapes this season, falls fatally under its sway the next, when all the rest of the country is clear. These and such circumstances as these, point as clearly as it is possible to point at the endemic source of cholera, although the *essential* cause of the disease cannot be ascertained.

But although we cannot in cholera, more than in periodic fever, ascertain the *essential* cause, there is no difficulty in showing the accidental auxiliary agents, which both in disposing to, and co-operating with it, give it much of its prevalence, and most of its fatal power. They consist in whatever deranges healthy action, and impairs constitutional vigour—such as unwholesome insufficient or irregular supplies of food—over labour—crowded and defective ventilation—dissolute habits, including vicious indulgences in intoxicating drink, and want of personal and domestic cleanliness—apprehension, anxiety, and inordinate emotions of the mind; and defective drainage, including sewerage, with resulting accumulation of organic matters. For the last, at least, the State and the authorities acting under it ought to be held responsible.

Whether the last named agent—defective drainage and its consequences—constitutes any thing positive to the *essential* cause of the disease, or is necessarily connected with it, or whether it only co-operates with those specified before it, in lowering the standard of health, by the production in excess of hydrogen and other gases, injurious to life, and thereby predisposes the body, through augmented susceptibility, for the action of the *essential* cause cannot be determined. It is pretty certain, however, that neither it in any quantity, nor any amount of the other agents classed with it, as disposing and co-operating powers, in the production and extension of cholera, can of themselves create it. If they could, not only part of Ireland, but of places nearer home, must have been decimated, some of them depopulated.

Whether excess or deficiency in the electric element, or disturbance in its ordinary relations, in the place where cholera appears, acts any part in the production of the disease, is not known. It seems probable that there is something abnormal in its distribution and movements; but that is all that can ever be reasonably conjectured, at least for the present, respecting it.

Whether there is any affinity between the *essential* cause of cholera and periodic fever, and, if any, of what kind, are questions that naturally suggest themselves in investigating the etiology of the former. That the question, as far as it relates to a certain affinity in the two cases, should be answered affirmatively, the following considerations seem to shew.

Swampy undrained soils, the banks of rivers, the margins of harbours, and other low lying localities, with places where organic remains are artificially accumulated, or allowed to accumulate, prove the most prolific positions in the production of cholera, as well as of periodic fever.

Both before the outbreak of cholera, and on its subsidence, fever of type, more or less distinctly marked, with predominance of gastric symptoms, are more frequent than usual.

Some remarks, as already stated, were made on this subject in the “Medical Notes on China;” it may be added that many circumstances might be cited, tending to show that there is a close connection, not only between the causes of cholera and periodic fever, but also between it and endemic fever generally.

All this is very vague and unsatisfactory, but with such generalities, negatives and probabilities, it is feared we must for the present at least, be as content as we can, unless we are disposed to leap to conclusions, without finding the steps of evidence by which alone they can be safely reached; or are willing to subscribe the creed, that cholera, however first produced, is reproduced only through the instrumentality of the human body—that, by whatever acts, combination of acts, or of accidents it began, it is continued solely by contagion.

The essential cause of cholera—the *causa sine qua non est*—as well as of other things in nature with what are more familiar, is, and perhaps ever will be, beyond the reach of human penetration; and were its nature and

properties ascertained, such knowledge might contribute little or nothing to our means of resisting its effects. But while this ignorance exists, and candour requires its confession, there is none respecting the associated concurrent agents, through whose influence it acquires extended prevalence and augmented fatal force. They have been already alluded to, and are so palpable as to be beyond question; and as they are on the surface, and susceptible of abatement or removal, it becomes us to deal with them promptly, vigorously, and perseveringly.

They divide themselves into two classes, the first of which concern private citizens, both as individuals, and as forming families; the second are subjects for legislative enactments, and consequent executive interference, constructive and restrictive.

Respecting the first, it may appear superfluous to reiterate what has been so often repeated, and universally admitted, namely the paramount importance of sobriety, cleanliness, and industry, in conjunction with sufficiency of nutritious food; nor to aver that if the former were more sedulously practised, there would be much less ground for complaint of the last than unhappily there is. It would be out of place to enlarge on such a subject here, but it may be allowable in passing to observe, that it behoves the more intelligent and wealthy members of the community, lay as well as clerical, to unite and help the more ignorant and needy in the great—vast as to results—reform, which it contemplates and embraces. A good deal has been attempted, and something has been done, but much is wanting. The work is great, requires many labourers, and gives scope for the combined exertions of enlightened philanthropists of all kinds. Yet we may say of it, without profanation or levity, and in the literal meaning of the words—truly the harvest is great but the labourers are few.

The second class of reformatory measures required for the object in view must devolve on the legislature, and on authorities constituted by it, as all experience shows they cannot be entrusted to the voluntary efforts of individuals, or to those of municipal, or other local self-acting bodies. The enlarged and disinterested consideration of men having authority, independent of the conflicting interests, and removed from the prejudiced opinions, affecting particular places, who look to the welfare of the whole, must do the work, if it be ever done effectually. Enlightened and

benevolent men in parliament and out of it have not been scared from the undertaking by the obstacles real and fictitious in the way. Let them however, especially those in parliament persevere, leading others to join them as they assuredly will, disregarding the selfish and ignorant objections raised against their truly patriotic efforts, and they cannot fail of their reward.

The measures especially required comprise the form and situation of houses, width of streets, complete underground drainage, and the instant removal or destruction of refuse organic substances from and on the surface, by strict enactments, rigidly enforced. This like the former, is a hacknied theme, but it not the less important on that account, and its interest can never be exhausted till the objects which it embraces are accomplished.

If it were objected that such interference by the state in the ordinary affairs of life would trench injuriously on the liberty of the subject, it should be answered that the functions of government are not merely repressive of public outrage, and punitive of offences against the person, that improvements of the people's condition at large ought to be its great aim, and that it can no more be justified in permitting the lieges, by omission of what is right or commission of what is wrong, to poison themselves on a great scale, than it would be in looking complacently on at the suicide of individuals.

It is true that the measures in question cannot be completed without much labour, and opposition, extending over uncertain periods of time, but they must ultimately become part of the hygienic economy of the land, if we would avoid the imputation of being criminally accessory to the death of thousands, and do our duty honestly to our neighbours. Unlike the other, which as they must be effected by individuals and families will depend on improvements in the intellectual and moral condition of the persons themselves, these can be achieved authoratively by the Magistrates. Both classes of reformatory measures are required to show what may yet be done for the welfare of the people, and how incalculably the people can contribute to it themselves, especially in respect of health; but while only one of them is at the command of power, that one should no longer be neglected. Had all been done which might have been done in this way, it may be asserted fearlessly that cholera would not have exerted the fatal

power which it did in 1832, and which it is exerting in 1849; and that other epidemics which have prevailed in the period would have had fewer victims.

Unsatisfactory and reproachful however as the sanitary condition of the country is, when it appears how much it might be improved by the moral, social and physical reforms alluded to above, it is excellent when compared with what it was at former periods of our history. On looking back to the terrible epidemics, which in the thirteenth, fourteenth, fifteenth, sixteenth, and even so late as the seventeenth century, under the names of pestilence, black death, sweating sickness, and plague ravaged this, and other portions of Europe, we find such was the destructive power of disease then, that the most sickly seasons recorded for nearly two-hundred years have been, in comparison with them, seasons of health and enjoyment. The questions which suggest themselves as to the cause of difference will be variously answered. Amid the obscurity which involves the subject, one thing however is evident, and seems to offer a sufficient explanation of the difference, namely, the miserable condition of the people in those remote periods relatively to the present, the results of ignorance, apathy, licence and oppression—periodical recurrence of famine—wretched habitations, wretchedly kept—and the total want of sewerage, with accumulations of filthy decomposing substances, producing corruption in every corner and at every turning, sufficient to poison every living thing in their neighbourhood.

Since those disastrous times much, especially of late, has been done to preserve the health of the people, but much remains to be done, and, looking at the spirit in which the subject is taken up, and the arguments furnished from without to illustrate and enforce it, especially the urgent ones supplied by the daily lists of death from cholera, much more it is hoped, will be done speedily. If the legislature executive, and people at large would co-operate heartily and systematically, each doing their proper part to further the work—if stagnant fluids on and near the surface, and poisonous exhalations from decomposing vegetable and animal matter were prevented—if the dwelling places of the poor were sufficiently large, properly ventilated and cleansed, while the inhabitants conducted themselves industriously and morally, the effect, though it might not realize expectation, would unquestionably be great. The sanitary state of the

people might then as far surpass that of the present time, as it does that of the fourteenth and fifteenth centuries: and this age in respect of health, would occupy a middle station between the dark ages which are past, and the practically enlightened age which is to come, and ought to come speedily. The *essential* cause of cholera, and of other febrile endemics might be brought into existence, but without indulging in idle visions, it may be predicted that they would be comparatively harmless, in as much as they would be deprived of the concurrent, fostering agencies, from which they derive their prevalence, and chief instruments of destruction. And not only would cholera and other endemic diseases, whether prevailing epidemically, or occurring rarely, be checked and mitigated, but other forms of disease would be lessened, while constitutional health would be invigorated. Such great ends are surely worthy of great means zealously employed.

POSTSCRIPT.

14th September, 1849.

IN addition to the 12 fatal cases of cholera recorded in the text, one occurred late on the 31st of August. The subject was received at 6 P.M. of that day, in a state of complete collapse, and died six hours after admission.

During the currency of the month there have been admitted 12 cases of cholera, two of which, being in a state of profound collapse, terminated speedily in death; 5 cases of bilious cholera which are doing well; 10 cases of febrile diarrhœa which are doing well; and one case of colic with spasms of the extremities.

The number of cases of cholera, and of allied affections, excluding fevers, and adopting the nomenclature proposed above, between the 4th July and this date, are as under.

CHOLERA ABILIOSA.			
Cases.	Cured	Dead.	Remain, coalescent or improving.
49	26	15	8
CHOLERA BILIOSA.			
25	19	0	6
FEBRILE DIARRHŒA.			
53	44	0	9
CONVULSIVE COLIC.			
11	10	0	1

Further experience has confirmed the opinion that the means specified at the commencement of this paper constitute fit and remedial treatment in the formidable division of cholera to which they especially refer; and established the conviction that when the impression is not overpoweringly severe at the onset, and when sinking has not made great progress, a large proportion of patients will pass safely through the disease, under their use, if perseveringly and unswervingly employed. Latterly the opiate draught administered on admission has been omitted, the treatment consisting almost exclusively of calomel, turpentine draughts, and turpentine enemata, with friction when required. A rubefacient, consisting of strong mercurial, and cantharides ointment, each one ounce, with half-an-ounce of oil of turpentine, well mixed, has been used with effect. The calomel, and turpentine draughts have generally been given every half hour, at first, in violent cases, and less frequently as the symptoms subsided, being suspended when tendency to collapse was overcome, or when bilious vomiting set in.

The practice in bilious cholera, febrile diarrhœa, and convulsive cholera, was simple, and similar in all, consisting of a bolus containing three grains of calomel, and one of crude opium, repeated every second, third, or fourth hour, according to the force of symptoms. Cathartics were often required in the colicky affections, as was occasionally abstraction of blood by venesection, or leeches. The remedies were extremely uniform, and have been uniformly successful.

It may be observed in conclusion that, up to this date, none of the nurses, medical attendants, or other persons associated with the sick of cholera have been affected by the disease, although one or more of the medical officers of the Establishment have been almost constantly in the ward, and one or other of the juniors has slept in an apartment close to it; and further that no case of cholera has arisen within the walls of the Hospital, while it has been frequently brought in, and had continual existence within them, upwards of ten weeks, amid a population of patients, servants, including the women who washed the foul linen, and officers with their families, of nearly 600 souls.

LEGG, PRINTER, GOSPORT.

FOOTNOTES.

[9] See postscript.

[17] An anonymous writer in the *Times* of the 11th August, states that Birmingham had like immunity in 1832, and up to that date of the present year.

[19] Paper by Dr. Mc William in the Medical Gazette of 15th June of this year.

*** END OF THE PROJECT GUTENBERG EBOOK TREATMENT OF
CHOLERA IN THE ROYAL HOSPITAL, HASLAR ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.